Cloning of Human TFEB and Analysis of Its Nutrient-Dependent Subcellular Localization in HeLa Cells

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Abstract

Transcription factor EB (TFEB) is a nutrient-responsive transcription factor that coordinates autophagy and lysosomal biogenesis. This proposal aims to construct an EGFP-tagged human TFEB expression plasmid and use it to visualize TFEB localization in HeLa cells under nutrient-replete, starvation, and mTOR-inhibition conditions. The central hypothesis is that nutrient deprivation or pharmacological inhibition of mTOR will increase nuclear accumulation of the reporter. The construct will be generated by PCR amplification of a defined human TFEB coding sequence, restriction-enzyme cloning into a mammalian expression vector, bacterial screening, and Sanger sequencing. Transfected cells will be evaluated by fluorescence microscopy and immunoblotting. All outcomes described here are expected results of a proposed experiment.

Keywords: TFEB; autophagy; lysosome; mTORC1; EGFP fusion protein; subcellular localization

1 Rationale and Research Basis

TFEB is a member of the MiT/TFE transcription-factor family and a major transcriptional regulator of the autophagy–lysosome pathway. Under nutrient-rich conditions, mTORC1-dependent phosphorylation favors cytoplasmic retention of TFEB. During nutrient starvation or mTOR inhibition, TFEB accumulates in the nucleus and activates genes involved in lysosomal biogenesis and cellular clearance [2, 3]. Nutrient-dependent control of nuclear export also contributes to TFEB distribution [1].

This project connects molecular cloning with a visible cell-biological phenotype. Its primary readout is the nuclear-to-cytoplasmic distribution of a fluorescent TFEB fusion. The study will test whether the sequence-verified reporter exhibits the expected nutrient-dependent redistribution in HeLa cells.

1.1 Research Objectives and Hypothesis

The first objective is to construct and sequence-verify an EGFP-tagged TFEB mammalian expression plasmid. The second is to compare its subcellular localization under normal culture, starvation, and mTOR-inhibitor conditions. The hypothesis is that starvation and mTOR inhibition increase the nuclear-to-cytoplasmic fluorescence ratio relative to the vehicle-treated control.

2 Experimental Design

2.1 Overall Workflow

The study proceeds from reference-sequence retrieval and primer design to PCR amplification, plasmid construction, clone verification, transient transfection, and localization analysis. Sequence verification is the gate between molecular cloning and cell-based experiments.

2.2 Construct Strategy and Materials

The starting outline proposes an N-terminal EGFP–TFEB fusion expressed from a CMV promoter, using pEGFP-C1 as a candidate backbone. The final design will specify a single backbone and a versioned TFEB coding sequence, then verify the promoter, fusion junction, multiple cloning site, bacterial replication origin, and bacterial and mammalian selection markers. The original outline's fusion orientation remains provisional until the sequence and literature checks are complete.

For an EGFP–TFEB construct with TFEB at the C terminus, the TFEB termination codon should be retained unless an additional downstream fusion is explicitly intended. For the alternative TFEB–EGFP orientation, the native TFEB termination codon must be omitted to permit translation into EGFP. Removing a stop codon does not by itself establish the correct reading frame; both junctions require sequence-level verification.

The planned molecular materials are a defined human TFEB template, the selected EGFP expression vector, high-fidelity DNA polymerase, two compatible restriction enzymes, DNA ligase, competent *Escherichia coli* DH5α, and media containing the antibiotic specified by the verified backbone. A circular plasmid map will show the final sequence features and cloning junctions.

2.3 Sequence Retrieval and Primer Design

A human TFEB reference transcript and coding sequence will be retrieved from NCBI with accession and version recorded. Each cloning primer will be reported in the $5'$ to $3'$ direction and annotated by protective bases, restriction site, any frame-adjustment or linker bases, and template-annealing sequence. The selected enzyme pair must cut the intended backbone at the required positions without cutting the TFEB insert internally. Primer length, melting temperature, and compatibility with the amplification conditions will be documented after the sequence is fixed.

2.4 PCR Amplification and Restriction Cloning

The TFEB insert will be amplified with a high-fidelity polymerase. A no-template control and an appropriate amplification control will accompany the reaction. The intended amplicon will be purified, and both insert and vector will be digested and purified before ligation. An initial insert-to-vector molar ratio of approximately 3:1 is proposed. The ligation will be transformed into competent DH5α cells, followed by selection using the backbone's confirmed antibiotic marker. The reaction composition, thermal program, digestion conditions, and transformation details remain to be specified for the selected reagents.

2.5 Clone Screening and Sequence Verification

Candidate colonies will be screened by colony PCR and diagnostic restriction digestion. Plasmid DNA from positive clones will be purified for Sanger sequencing with vector-flanking and, where necessary, internal primers. A construct will be accepted only when the complete TFEB insert, both junctions,

fusion reading frame, and terminal stop codon match the intended sequence without unintended substitutions, insertions, or deletions. An expected-size band alone will not establish sequence identity.

2.6 Cell Culture, Transfection, and Treatment

Sequence-verified plasmid will be transiently transfected into HeLa cells. The proposed comparison includes normal culture with vehicle, starvation medium, and Torin1 treatment. The final methods will identify the culture medium, transfection reagent, cell density, DNA amount, and treatment concentration and duration. Empty EGFP vector and mock-transfected cells will provide expression and background controls. A matching vehicle control will accompany Torin1 treatment.

2.7 Imaging and Localization Analysis

After treatment, cells will be fixed and counterstained with DAPI. Fluorescence images will be collected using consistent acquisition settings. Nuclear and cytoplasmic regions will be defined for each eligible transfected cell, and background-corrected mean fluorescence will be used to calculate the nuclear-to-cytoplasmic ratio:

$$R_{\mathrm{N/C}} = \frac{I_{\mathrm{nucleus}} - I_{\mathrm{background}}}{I_{\mathrm{cytoplasm}} - I_{\mathrm{background}}}. \tag{1}$$

At least three independent experiments are planned. The number of cells per replicate, eligibility criteria, and handling of background-level cytoplasmic signals will be specified before analysis. Cells within one experiment will not be treated as independent biological replicates. The final statistical comparison will account for the experimental unit and the planned control-versus-treatment comparisons.

2.8 Expression Validation

An anti-GFP Western blot will assess fusion-protein expression and possible cleavage products. The expected molecular mass will be calculated from the finalized construct. The final protocol will specify protein extraction, loading control, antibody information, and interpretation of additional bands. Expression validation will be considered alongside the localization data.

3 Expected Results and Criteria for Success

3.1 Molecular Cloning

PCR is expected to produce an amplicon matching the finalized TFEB insert design, with no corresponding product in the no-template control. Colony PCR and diagnostic digestion should yield the predicted fragment pattern. Exact amplicon and restriction-fragment lengths will be recorded with the completed sequence design.

The decisive criterion is Sanger confirmation of the complete intended fusion construct. Successful cloning requires correct insert identity and orientation, both junctions, the continuous open reading frame, and the intended termination codon. The final report will include a plasmid map, a cloning workflow, and explicitly labeled schematic predictions for the gel patterns.

3.2 Expression and Nutrient-Dependent Localization

The reporter is expected to show a higher nuclear-to-cytoplasmic fluorescence ratio after starvation or Torin1 treatment than in the vehicle-treated condition. An anti-GFP-reactive band consistent with the finalized fusion-protein size would support expression of the intended product.

Evidence for nutrient responsiveness will require a reproducible treatment effect across independent experiments, supported by expression and imaging controls. The magnitude and uncertainty of the localization change will be reported alongside the statistical comparison. A failure to observe redistribution would prompt checks of sequence, fusion orientation, expression level, treatment response, and imaging conditions.

4 Discussion and Limitations

The proposed design links molecular construction to a testable localization phenotype. Its main limitation is that overexpression and fluorescent tagging can influence TFEB regulation. Fusion orientation therefore requires explicit justification, and a reporter response should not be equated with a complete description of endogenous TFEB behavior.

The localization assay examines redistribution rather than autophagic flux or lysosomal function directly. Changes in nuclear localization alone will not establish increased autophagic degradation. Follow-up studies could compare reporter localization with endogenous TFEB or test selected downstream targets. These extensions are outside the initial cloning-and-localization objective.

Author Contributions

Siyuan Jiang will lead TFEB sequence retrieval, vector selection, cloning and primer design, molecular methods, plasmid and workflow diagrams, clone-verification criteria, and the related literature review. Ziyan Song will lead the abstract, biological rationale, HeLa-cell experiments, localization and expression analysis, cell-related expected results, discussion, and the related literature review. Both authors will review the construct-to-cell-assay interface and the integrated report.

AI Use Declaration

AI assistance was used to organize the supplied outline, identify preliminary reference and construct-logic issues, and prepare the LaTeX manuscript structure. The authors will review the final scientific content, sequence design, experimental parameters, references, and contribution statement before submission.

References

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